The Campus of Tomorrow

Higher Colleges of Technology

كليات التقنية العليا

CDS 2413- PROGRAMMING FOR DATA SCIENCE

Week 3-4

Week 3-4	CLO 2: Apply data analysis and visualization techniques to support data-driven decision-making

Document Revision CONTROL (DRC)

Version	Author	Effective Date	Change Description	DRC No
1.0	Dr. Mohammed Abdul Rahim	2/05/2023	Define the first version	001
	Dr. Amala Rajan		Review the first version	

X.Y
X: Major change
Y: minor change

Lecture Outline

Descriptive Statistics

Types of Descriptive Statistics: Measures of Frequency, Central Tendency, Dispersion and Position

Python Programming for Descriptive Statistics

LECTURE OBJECTIVES

- By the end of this course you will be able to
  - Demonstrate the understanding of different types of descriptive statistics and their benefits
  - Write python programming to analyze the given data using descriptive statistics

Descriptive Statistics

Descriptive Statistics

- Descriptive statistics summarize and describe data.
- They provide an overview but do not allow for conclusions.
- Types include:
  - Descriptive Statistics
  - Inferential Statistics
- Descriptive statistics help us understand data by summarizing it
- Inferential statistics allow us to make predictions or inferences.

Benefits of Descriptive Statistics

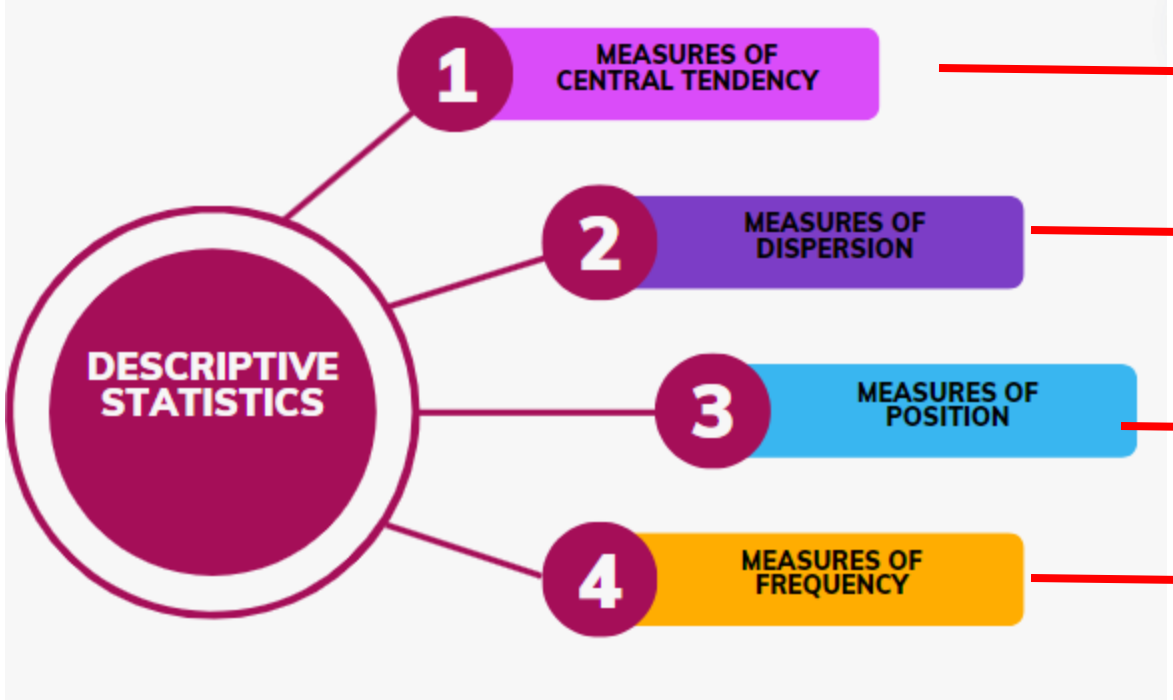
- They can help you to
  - Understand your data better.
  - Communicate your findings to others.
  - Identify outliers to avoid drawing incorrect conclusions about your data.
  - Compare different groups or variables.
  - Simplify data for visualization.

Types of Descriptive Statistics

- **Types of Descriptive Statistics**

1. Measures of Central Tendency
2. Measures of Dispersion
3. Measures of Position
4. Measures of Frequency

Descriptive Analysis - Types



Mean, Median, Mode

Range, Variance and Std Deviation

Skewness, Quantiles

Count, Frequency

Measures of Central Tendency

- Measures of central tendency provide a central value around which data clusters.
- Central tendency measures include:
- **Mean:** Average value.
- **Median:** Middle value.
- **Mode:** Most frequent value.

- Examples of Central Tendency

1. **Mean:** Average salary in a company.
2. **Median:** Median height of a population.
3. **Mode:** Most common crime rate in a city.

These examples illustrate how central tendency measures are used in real-world scenarios.

Measures of Central Tendency – Case Study

- HCT wants to assess the performance of students in FWA around various department
  - Perform descriptive statistics for Final Column
  - You have to use Marks.csv

ID	Division	Quiz	Practical	Project	Final
H00247	EDU	84	67	69.6	67
H00258	HLT	74	68	67	73
H00269	ENG	88	86	75.6	74
H00224	EDU	74	71	65.8	71
H00269	EDU	22	65	66.8	78
H00277	ENG	40	68	76.4	68
H00277	HLT	71	77	64.6	70
H00293	BUS	50	74	60.6	78
H00247	CIS	61	70	66.4	82

Note: Descriptive statistics can be applied for only one column

Measures of Central Tendency - Python

```
# find the average of each column
ds.mean()
```

```
# find the average of Final marks
ds['Final'].mean()
```

```
# find the averge marks obtained by each students - mean rowise
ds.mean(axis=1)
```

Measures of Central Tendency - Python

```
#find the middle mark of the all marks
ds.median()
```

```
#find the middle mark of the Project marks
ds['Project'].median()
```

```
# Which department have more entries
ds['Division'].mode()[0]
```

Measures of Dispersion

- Measures of dispersion show how spread out the data values are around the central tendency.
- Dispersion measures include:
- **Variance:** Spread of data values.
- **Standard Deviation:** Square root of variance.
- **Interquartile Range (IQR):** Range of middle 50% of data.

- Examples of Dispersion

1. **Variance:** Distribution of employee salaries.
2. **Standard Deviation:** Height differences in a population.
3. **IQR:** Crime rates in different neighborhoods.

- These examples demonstrate the application of dispersion measures in various contexts.

Measures of Dispersion- Python

```
# Find the minimum marks scored in the final
ds['Final'].min()
```

```
# Find the maximum marks scored in the final
ds['Final'].max()
```

```
# Find the range of marks scored in the final
final_range= ds['Final'].max() - ds['Final'].min()
print(final_range)
```

Measures of Dispersion- Python

```
# Find how finals marks are deviated from the average
ds['Final'].std()
```

```
#Find the variance of the final marks - square of std dev
ds['Final'].var()
```

Measures of Position

- Position measures indicate the relative standing of data values within a dataset.
- Position measures include:
- **Quartiles:** Divide data into four equal parts.
- **Skewness:** Asymmetry of the data distribution.

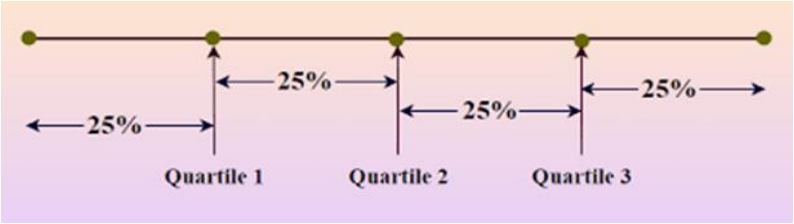
- Examples of Position Measure

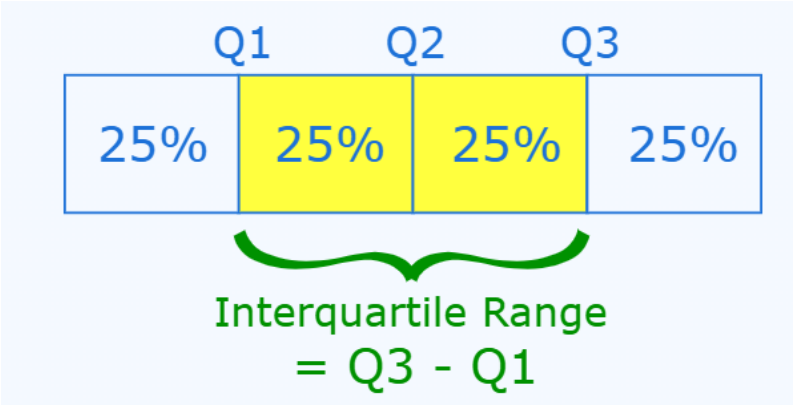
1. **Quartiles:** Distribution of student marks.
2. **Skewness:** Symmetry of data distribution.

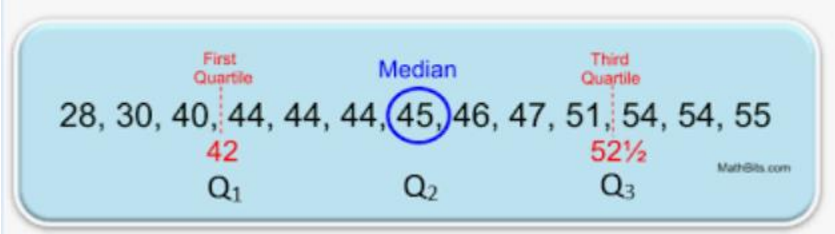
- These examples show how position measures are used to understand data distribution.

Measures of Position

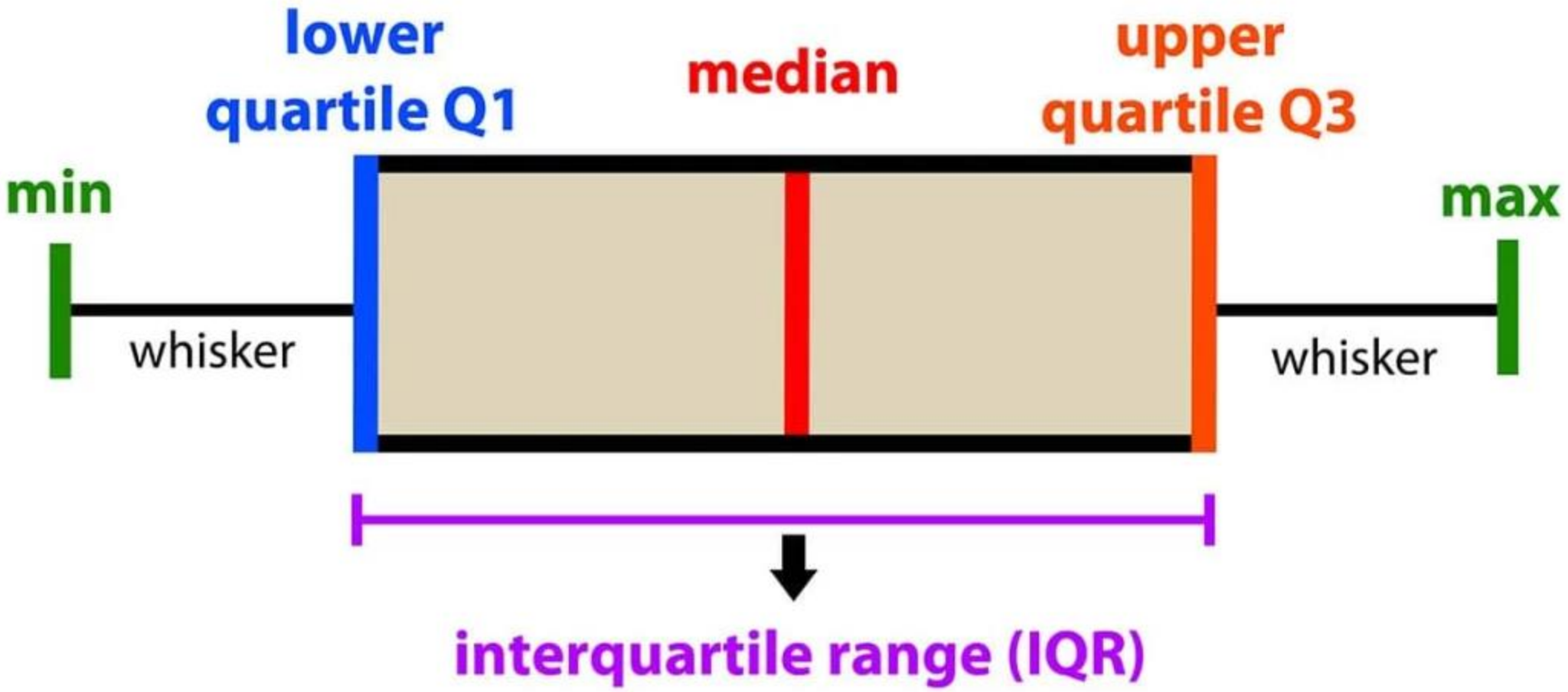
- The most common being quartiles and skewness.
  - **Quartiles**: Quartiles divide a data set into four equal parts.
    - The **First Quartile** (Q1) is the middle value of the bottom half of the data set,
    - The **Second Quartile** (Q2) is the median,
    - The **Third Quartile** (Q3) is the middle value of the top half of the data set, and
    - The **Fourth Quartile** (Q4) is the largest value in the data set.





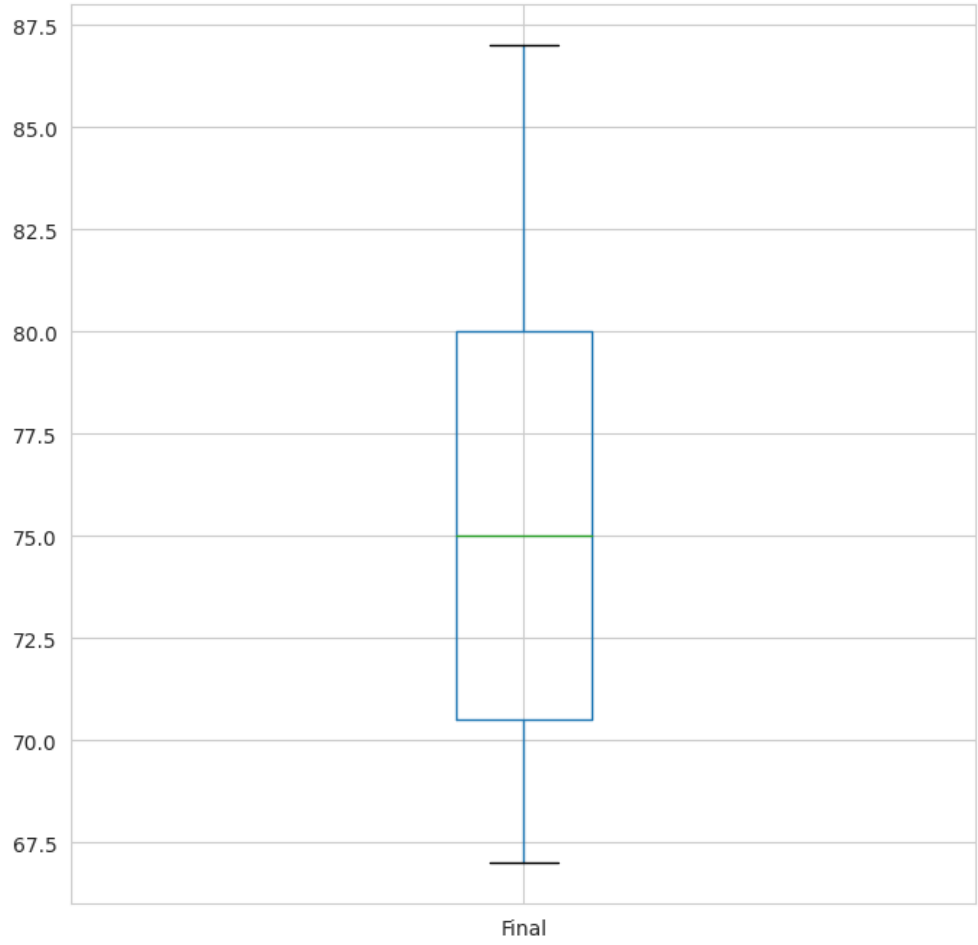


Measures of Position



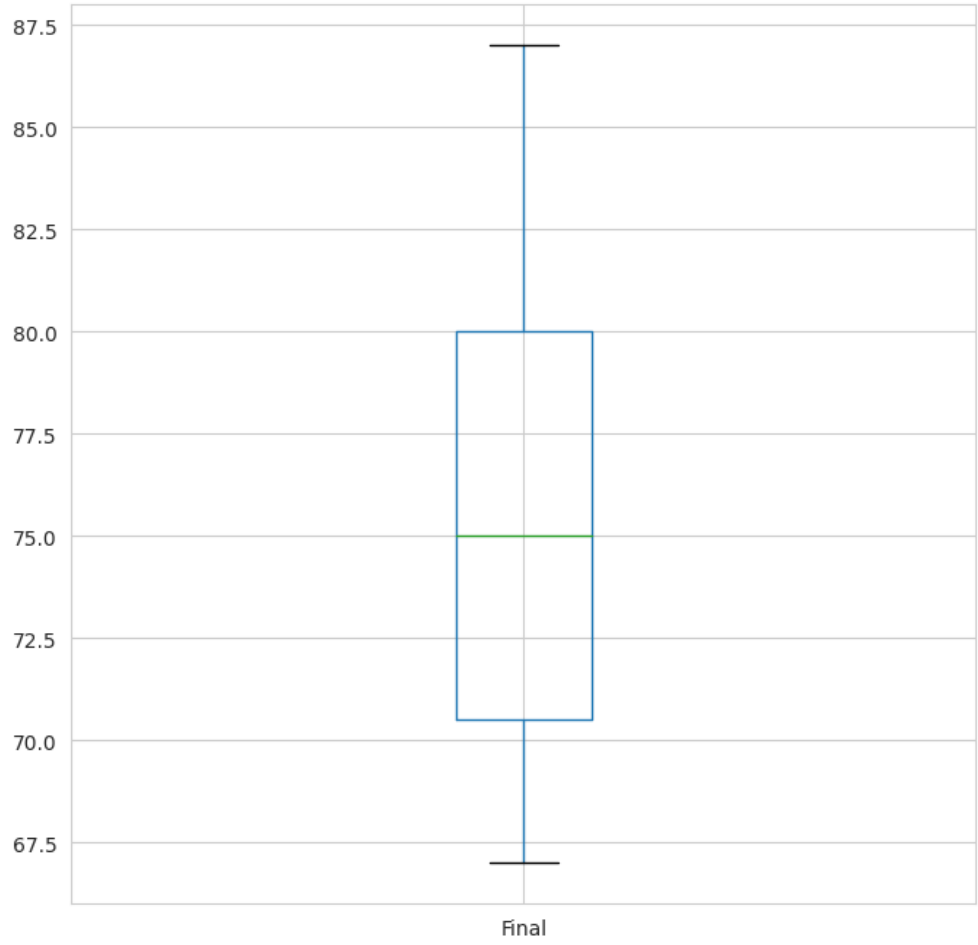
Measures of Position

- From the graph, find
  - Minimum
  - Maximum
  - First Quartile
  - Second Quartile
  - Third Quartile
  - Median
  - IQR



Measures of Position

- From the graph, find
  - Minimum - 67
  - Maximum - 87
  - First Quartile - 69
  - Second Quartile - 75
  - Third Quartile - 80
  - Median – 75
  - IQR – 80 – 69 = 11



Measures of Position – Case Study

- **Quartile**
- HCT is interested in understanding the distribution of FWA marks for its students.
- HCT divides the FWA marks into four equal groups, or quartiles.
- The first quartile (Q1) represents the FWA of the bottom 25% of students.
- The second quartile (Q2) represents the FWA of the bottom 50% of students.
- The third quartile (Q3) represents the FWA of the bottom 75% of students.
- The fourth quartile (Q4) represents the FWA of the bottom 100% of students

Measures of Position- Python

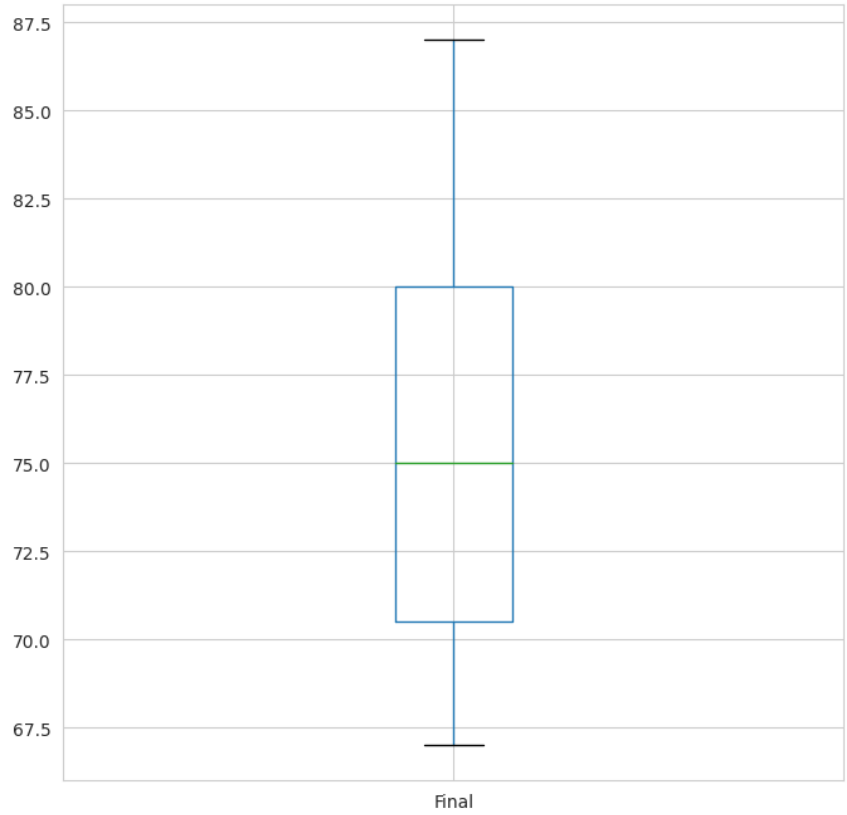
```
# find the different quantiles for the final marks
five_num = [ds["Final"].quantile(0),
            ds["Final"].quantile(0.25),
            ds["Final"].quantile(0.50),
            ds["Final"].quantile(0.75),
            ds["Final"].quantile(1)]

five_num
```

```
# find the inter quartile range(IQR) for final marks
ds["Final"].quantile(0.75) - ds["Final"].quantile(0.25)
```

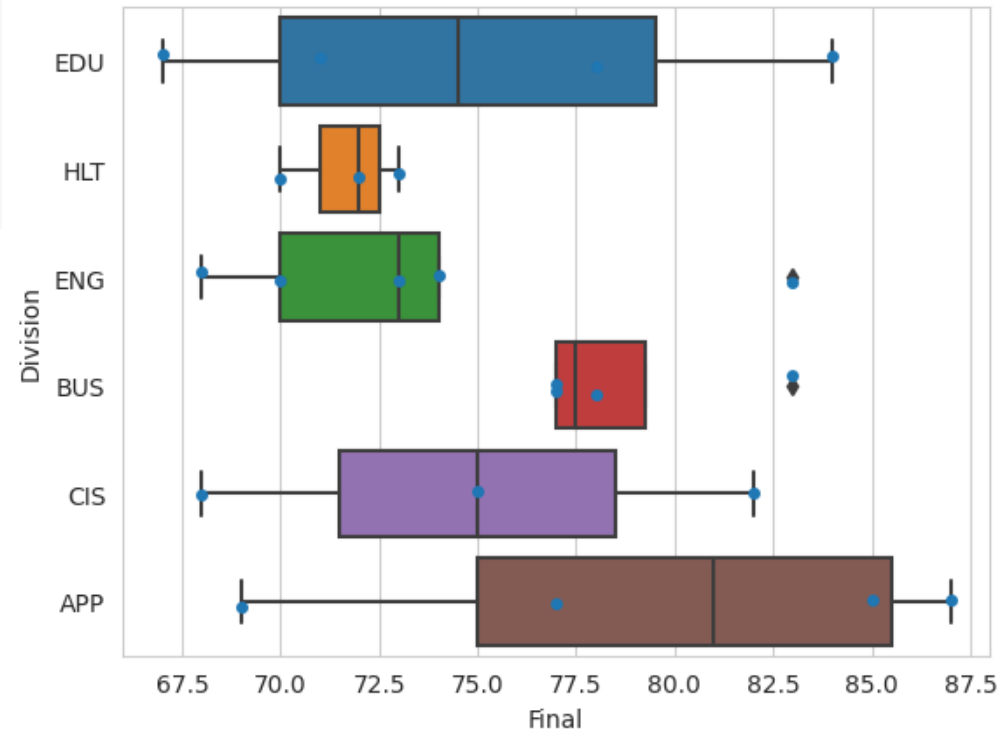
Measures of Position- Python – mat plot

```
# Draw the box plot for quartiles of final marks using mat plot library
import matplotlib.pyplot as plt
ds.boxplot(column="Final",
                return_type='axes',
                figsize=(8,8))
```



Measures of Position- Python – sea born

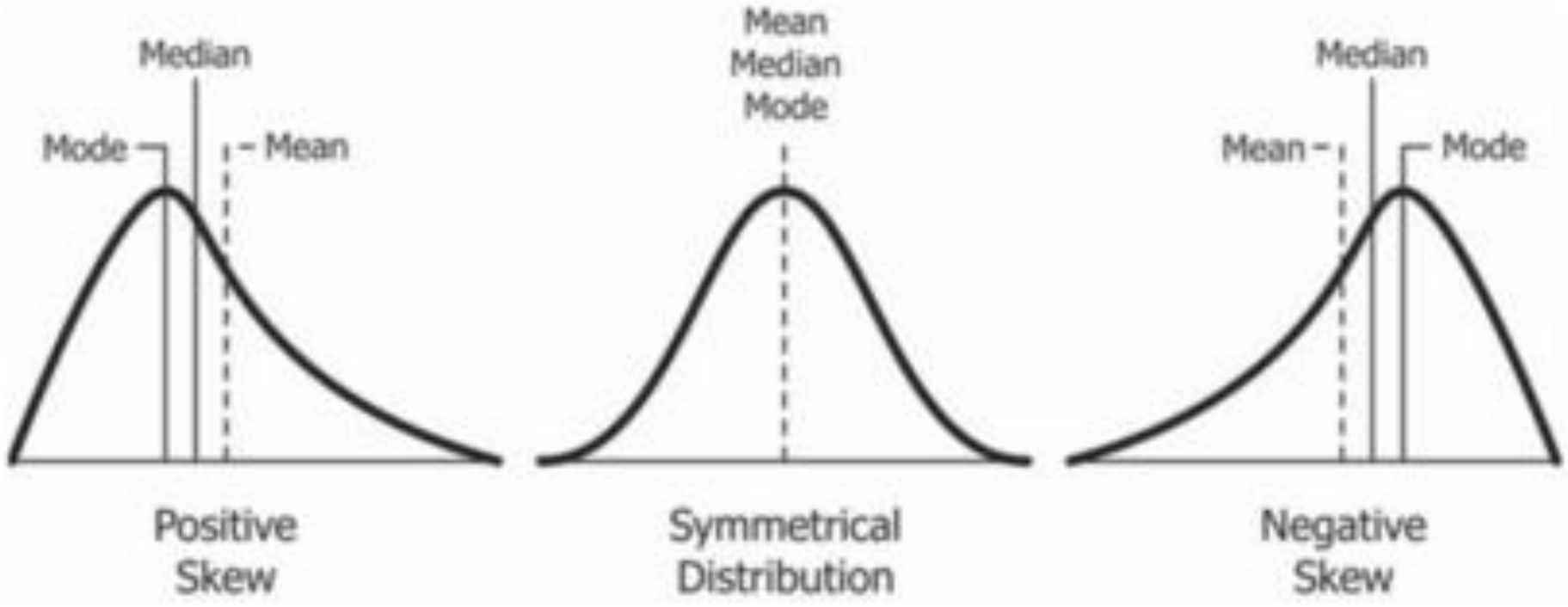
```
# Draw the box plot for quartiles of final marks departmentwise using sea born library
import seaborn as sns
sns.set_style('whitegrid')
ax= sns.boxplot(x='Final',y='Division',data=ds)
ax = sns.stripplot(x="Final", y="Division",data=ds)
```



Measures of Position- Skewness

- **Skewness** is **a measure of the asymmetry of a distribution.**
- The skewness of a distribution can be calculated using the following formula:
    - **Skewness = (Mean - Median) / Standard deviation**
- If the coefficient of skewness is equal to 0 or approximately close to 0 or 0, then the graph is said to be symmetric and data is normally distributed.
- If the coefficient of skewness is greater than 0, then the graph is said to be positively skewed with the majority of data values less than mean.
- If the coefficient of skewness is less than 0, then the graph is said to be negatively skewed with the majority of data values greater than mean.

Measures of Position - Skewness



Measures of Position- Python

```
# check the skewness of final marks
ds['Final'].skew()
```

Measures of Frequency

- Frequency measures describe how often values occur in a dataset.
- Frequency measures include:
- **Frequency Table:** List of values and their frequencies.
- **Histogram:** Bar chart showing frequency distribution.
- **Pie Chart:** Circle divided into sectors showing proportions.

Measures of Frequency - Python

- There are two types of frequency:
- Absolute frequency is the number of times a particular value occurs in a set of data.
    - For example, if the number of students studying in grade 5 is 4, then the absolute frequency of 5 is 4.
- Relative frequency is the proportion of times a particular value occurs in a set of data. It is calculated by dividing the absolute frequency by the total number of values in the data set.
    - For example, if the number of students studying in grade 5 is 4 out of 16 students(total), then the relative frequency of 5 is 4/16=0.25.

Grade	Frequency	Relative frequency
1	2	$\frac{2}{16}$ = 0.125 = 12,5%
2	2	$\frac{2}{16}$ = 0.125 = 12,5%
3	4	$\frac{4}{16}$ = 0.250 = 25%
4	2	$\frac{2}{16}$ = 0.125 = 12,5%
5	4	$\frac{4}{16}$ = 0.250 = 25%
6	2	$\frac{2}{16}$ = 0.125 = 12,5%

Measures of Frequency - Python

```
# find the number of students
ds['ID'].count()
```

```
23
```

```
# find the number of students division wise - absolute frequency
ds['Division'].value_counts()
```

```
ENG    5
EDU    4
BUS    4
APP    4
HLT    3
CIS    3
```

```
# find the number of students division wise with repsect to total students - relative
ds['Division'].value_counts(normalize=True)
```

```
ENG    0.217391
EDU    0.173913
BUS    0.173913
APP    0.173913
HLT    0.130435
CIS    0.130435
Name: Division, dtype: float64
```

5/23=0.217391

Descriptive Statistics

```
# Descriptive stat for all columns
ds.describe().T
```

Transpose

	count	mean	std	min	25%	50%	75%	max
Quiz	23.0	65.782609	17.549816	22.0	60.0	70.0	75.5	97.0
Practical	23.0	73.652174	10.503717	60.0	65.0	71.0	80.0	98.0
Project	23.0	71.086957	7.052228	60.6	66.1	69.6	76.4	85.2
Final	23.0	75.695652	6.041032	67.0	70.5	75.0	80.0	87.0

Descriptive Statistics

```
# Descriptive stat for categorical columns
ds.describe(include=object).T
```

	count	unique	top	freq
ID	23	11	H00293	6
Division	23	6	ENG	5

Categorical Column

Solve and Employing Descriptive Statistics – Case Study

Descriptive Statistics - Function

- HCT wants to assess the distribution of marks for each exam
- Create a general function to calculate the descriptive statistics for the given exam
  - The parameter of the function should be
    - Dataset
    - Field to which descriptive stat should be calculated

Descriptive Statistics - Function

```python
def Desc_stat(ds,var):
  mean=ds[var].mean()
  median=ds[var].median()
  mode=ds[var].mode()
  minimum=ds[var].min()
  maximum=ds[var].max()
  range=ds[var].max() - ds[var].min()
  std_dev=ds[var].std()
  variance=ds[var].var()
  skew=ds[var].skew()
  kurtosis=ds[var].kurt()
  count=ds[var].count()
  Quart = [
            ds[var].quantile(0),
            ds[var].quantile(0.25),
            ds[var].quantile(0.50),
            ds[var].quantile(0.75),
            ds[var].quantile(1),
            ds[var].quantile(0.75) - ds[var].quantile(0.25)
          ]
```

Dataset

Field to find the descriptive statistics

```python
summary = {
            "Average ": mean,
            "Median ": median,
            "Mode ": mode,
            "Minimum ": minimum,
            "Maximim ": maximum,
            "Range ": range,
            "Std Dev ": std_dev,
            "Variance ": variance,
            "Skewness ": skew,
            "Kurtosis ": kurtosis,
            "Minimum ": Quart[0],
            "25th Pecentile ": Quart[1],
            "50th Pecentile ": Quart[2],
            "75th Pecentile ": Quart[3],
            "Maximum ": Quart[4],
            "Inter Quartile Range ": Quart[5],
            "Count ": count
            }
return summary
```

Descriptive Statistics - Function

- Output of the function

```
Desc_stat(ds,'Project')
```

```
{'Average ': 71.08695652173914,
 'Median ': 69.6,
 'Mode ': 0    64.6
 1    71.0
 2    76.4
 3    81.4
 Name: Project, dtype: float64,
 'Minimum ': 60.6,
 'Maximim ': 85.2,
 'Range ': 24.6,
 'Std Dev ': 7.052227523518953,
 'Variance ': 49.73391304347827,
 'Skewness ': 0.4031063757141921,
 'Kurtosis ': -0.8503499013990585,
 '25th Pecentile ': 66.1,
 '50th Pecentile ': 69.6,
 '75th Pecentile ': 76.4,
 'Maximum ': 85.2,
 'Inter Quartile Range ': 10.300000000000011,
 'Count ': 23}
```

Dataset

Field to find the descriptive statistics

Descriptive Statistics - Quiz

1. What is the median score for the Quiz?

2. What percentile of students scored less than 65 in the Practical?

3. Is a Project score of 70 within the interquartile range (IQR) for the Project scores?

4. What is the interquartile range (IQR) for the Final scores?

5. How many students scored above 80 in the Practical?

	count	mean	std	min	25%	50%	75%	max
Quiz	22	65.409	17.869	22	60	68.5	75.75	97
Practical	23	73.652	10.504	60	65	71	80	98
Project	23	71.087	7.0522	60.6	66.1	69.6	76.4	85.2
Final	23	75.696	6.041	67	70.5	75	80	87

Descriptive Statistics - Quiz

1. What is the maximum score in the Project column?

2. What range of scores defines 'middle-performing' students for the Quiz?

3. What percentage of students scored below 60 in the Quiz?

4. If a student's Final score is 87, is this the highest score achieved?

5. What is the mean and standard deviation of the Practical scores?

	count	mean	std	min	25%	50%	75%	max
Quiz	22	65.409	17.869	22	60	68.5	75.75	97
Practical	23	73.652	10.504	60	65	71	80	98
Project	23	71.087	7.0522	60.6	66.1	69.6	76.4	85.2
Final	23	75.696	6.041	67	70.5	75	80	87

Case Study – Data Analysis-Critical Analysis

1. What is the total number of students who took the Quiz, Practical, Project, and Final assessments?

2. Calculate the range of scores for the Quiz.

3. How does the mean of Practical scores compare to the mean of Project scores? Which is higher?

4. What is the z-score for a student who scored 80 in the Final exam?

5. If a student scored 90 in the Quiz, what percentile would they approximately belong to?

	count	mean	std	min	25%	50%	75%	max
Quiz	22	65.409	17.869	22	60	68.5	75.75	97
Practical	23	73.652	10.504	60	65	71	80	98
Project	23	71.087	7.0522	60.6	66.1	69.6	76.4	85.2
Final	23	75.696	6.041	67	70.5	75	80	87

Case Study – Data Analysis-Critical Analysis

1. What is the difference between the 75th percentile and the mean for Practical scores?
2. For the Project scores, what proportion of students scored below 66.1?
3. If a student scored 60 in the Practical and 60.6 in the Project, are they below average in both?
4. What is the combined mean of all four assessments (Quiz, Practical, Project, Final)?
5. How many points are needed to move from the median to the 75th percentile in the Final scores?

	count	mean	std	min	25%	50%	75%	max
Quiz	22	65.409	17.869	22	60	68.5	75.75	97
Practical	23	73.652	10.504	60	65	71	80	98
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Final	23	75.696	6.041	67	70.5	75	80	87

Case Study – Descriptive Statistical Analysis Report

- Report should contain:
  1. Analysis be category (Quiz, Practical, Project, and Final)
  2. Comparative Insights – What it tells
  3. Key Recommendations – What you recommend based on the insights
  4. Conclusion – summary of all with limitation and future directions

	count	mean	std	min	25%	50%	75%	max
Quiz	22	65.409	17.869	22	60	68.5	75.75	97
Practical	23	73.652	10.504	60	65	71	80	98
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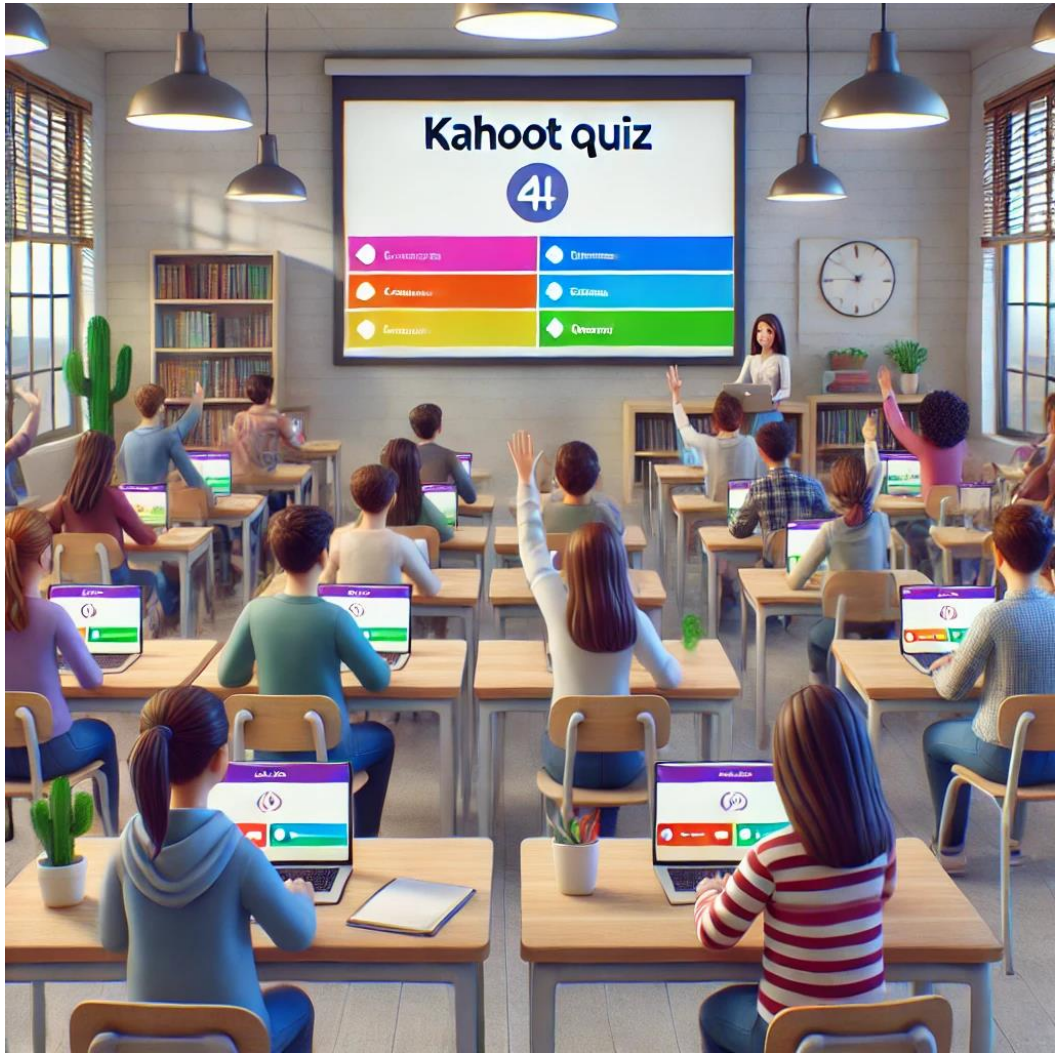
Final	23	75.696	6.041	67	70.5	75	80	87

Project

6.	Create a script to develop a Python function for descriptive statistics. The input for the function should be the sample and the field to perform the descriptive statistics.	1
7.	Create a program to random sampling of size 150 and find the descriptive statistics for the dependent variable from the sample [Apply the descriptive function which you created].	1
8.	Create a script for systematic sampling by giving certain conditions and finding the desc stat for the dependent variable from the sample [Apply the descriptive function which you created].	1
9.	Create a detailed descriptive statistics report about the dependent variable of the chosen dataset.	1

Kahoot Time!



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Thank You